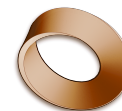


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



1.2 – Hooke's Law

Unit 1: Oscillations & Waves

The question: *can we predict how far a spring stretches before we pull it?* Last lesson you learned to **model** — you drew the pendulum and made the physics visible on paper. Today the model is the starting point, not the finish. You will draw the spring in front of you, then go and **collect the data** that turns your drawing into a rule you can trust: measure, tabulate, plot. Nobody is going to tell you the answer — your data is the answer.

Part 1 — Model the spring

1) **Draw** the *same* spring in each of the three lanes: with no mass on it, with one mass, and with three masses. The ceiling and the dashed line are printed for you — everything hanging below them is yours to draw. Show the spring, the hanger and the masses, and the force arrows acting on the mass, each one labelled.

The dashed line is the **natural length** — where the bottom of the spring sits with nothing on it. Every extension you measure today is counted from that line, not from the ceiling and not from the bench. Mark on your drawing *exactly* where on the spring you will read from, and read from the same place every time.

①	②	③
no mass	one mass	three masses
extension = _____ cm	extension = _____ cm	extension = _____ cm

Name: _____

Date: _____

Part 2 — Variables

Every investigation has three kinds of variable. Name yours before you touch the apparatus.

2) **Independent** — the one you **change on purpose**, one step at a time. What will you change?

3) **Dependent** — the one you **measure** each time you change the independent variable. What will you measure?

4) **Control** — the ones you **keep the same** so the test stays fair. Name two, and say why each one would spoil your results if it changed.

Name: _____

Date: _____

Part 3 — Collecting the data

Method

1. Measure the spring's **natural length** — nothing hanging on it — and write it above its table.
2. Add masses **one at a time**; each 100 g pulls with about **1 N**. Measure the **total length** each time and work out **extension** = **total length** – **natural length** straight away.
3. Six readings, then **change spring**. Row 1 is your check — no mass, no extension.

Stop adding masses if the coils stay apart.

Safety: eye protection on; feet clear of the masses.

Spring 1 natural length = _____ cm

Mass added / g	Force / N	Total length / cm	Extension / cm
0			

Spring 2 natural length = _____ cm

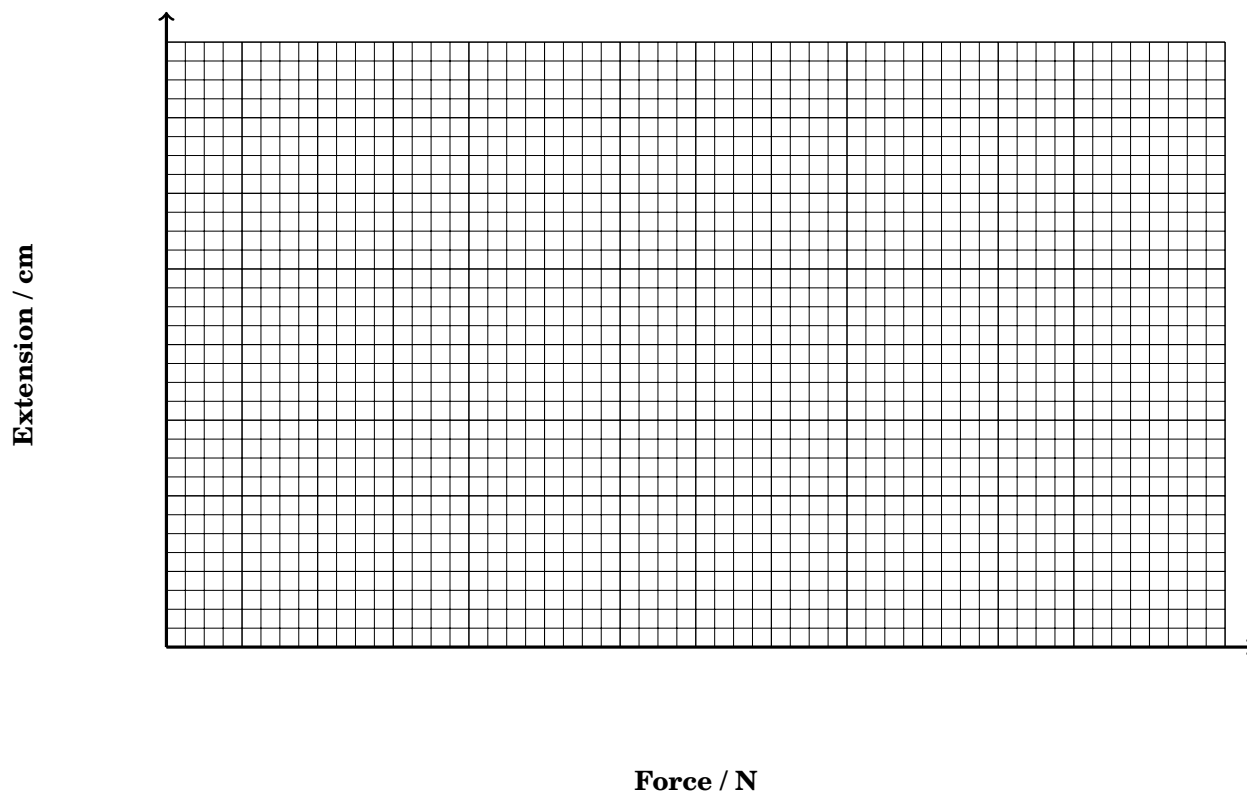
Mass added / g	Force / N	Total length / cm	Extension / cm
0			

Spring 3 natural length = _____ cm

Mass added / g	Force / N	Total length / cm	Extension / cm
0			

Part 4 — The graph

5) **Plot all three springs on the same grid** — force on the horizontal axis, extension on the vertical axis. Choose scales that use most of the grid and write your values along each axis. Use a different mark for each spring ($\times$, $\bullet$, $+$) and label them 1, 2, 3, then draw a straight **line of best fit** through each set with a ruler — do not join the dots.



Extension

Your springs are a Pasco *Equal Length Spring Set*. The maker states that each spring is **25, 30, 35, 40** or **50 N/m**, accurate to within $\pm 5\%$.

Does your data prove it?

Part 5 — Homework: the same model, built by a machine

HOMEWORK

You built a line of best fit by eye. Now make a spreadsheet build one from the same numbers, and see whether the machine agrees with you.

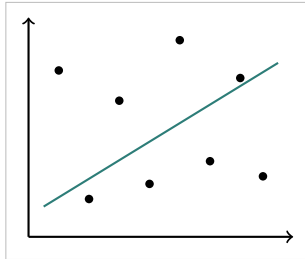
1. Open a new spreadsheet. Type two column headings: Force (N) and Extension (cm), and type in the readings for **one** of your three springs.
2. Select both columns and **insert a chart**. Choose a **scatter** chart — points only, no joining lines.
3. Add a **trendline** — the spreadsheet's name for a line of best fit.
4. Write one sentence: does the trendline agree with the line you drew by hand? If it does not, which do you trust, and why?

Hand in: the chart, with your sentence underneath it.

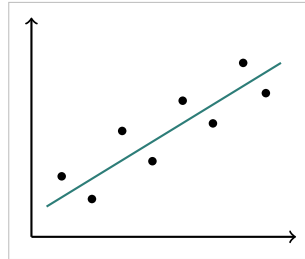
Part 6 — Conclusion

Describe: Give a detailed account or picture of a situation, event, pattern or process.

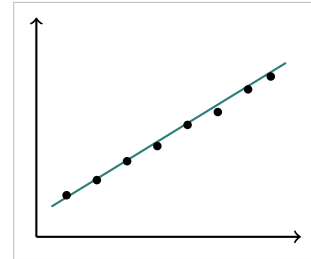
Correlation is how closely the points follow the line. It is not the same as the steepness — a gentle line can be followed very closely, and a steep one can be a mess. These three sets of data all have a line through them; only the last one has the points *sticking to it*.



weak



moderate



very strong

6) **Describe** the mathematical relationship between force and extension for an ideal spring. Say what *shape* the relationship is, what happens to the extension when the force doubles, and how strong the correlation is — use your own graph as the evidence.

7) **Predict** the extension for a force you did *not* test, using your line of best fit. Show on the graph how you read it off.

8) **Compare** your three lines. What is the same about all three, and what is different? What does the *steepness* of a line tell you about that spring?

How this is marked — Criterion C

This investigation is assessed against **Criterion C: Processing and evaluating**, out of **8**. This is the whole criterion, word for word from the MYP Sciences guide. **Read it before you write your conclusion, not after.**

By the end of Year 3, you should be able to:

- i. present collected and transformed data
- ii. interpret data and describe results using scientific reasoning
- iii. discuss the validity of a hypothesis based on the outcome of the scientific investigation
- iv. discuss the validity of the method
- v. describe improvements or extensions to the method.

Level	Level descriptor
0	The student does not reach a standard identified by any of the descriptors below.
1–2	The student is able to: i. collect and present data in numerical and/or visual forms ii. interpret data iii. state the validity of a hypothesis with limited reference to a scientific investigation iv. state the validity of the method with limited reference to a scientific investigation v. state limited improvements or extensions to the method.
3–4	The student is able to: i. correctly collect and present data in numerical and/or visual forms ii. accurately interpret data and describe results iii. state the validity of a hypothesis based on the outcome of a scientific investigation iv. state the validity of the method based on the outcome of a scientific investigation v. state improvements or extensions to the method that would benefit the scientific investigation.
5–6	The student is able to: i. correctly collect, organize and present data in numerical and/or visual forms ii. accurately interpret data and describe results using scientific reasoning iii. outline the validity of a hypothesis based on the outcome of a scientific investigation iv. outline the validity of the method based on the outcome of a scientific investigation v. outline improvements or extensions to the method that would benefit the scientific investigation.
7–8	The student is able to: i. correctly collect, organize, transform and present data in numerical and/or visual forms ii. accurately interpret data and describe results using correct scientific reasoning iii. discuss the validity of a hypothesis based on the outcome of a scientific investigation iv. discuss the validity of the method based on the outcome of a scientific investigation v. describe improvements or extensions to the method that would benefit the scientific investigation.

Where each strand lives in this lab. **i** — your three data tables in Part 3, and the graph in Part 4. **ii** — what you wrote about the shape of the lines and what steepness means. **iii** — whether your data supports the rule you predicted. **iv** — whether the way you measured could be trusted. **v** — what you would change if you did it again.

The bold words are the whole difference between the bands. Every band says the same five things; what changes is how far you take them — from *state* to *outline* to *discuss*.